Reinforcement Learning (AI3000)

Lecture 02

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Markov Decision Process (MDP)

- An MDP is a generalization of a Markov process (or Markov chain) with the following elements:
  - State space $\mathcal{S}$
  - Action space $\mathcal{A}$
  - Reward alphabet $\mathcal{R} \subseteq \mathbb{R}$
- A process $\{(S_n)_{n\geq 0}, (A_n)_{n\geq 0}, (R_n)_{n\geq 1}\}$ is called a Markov decision process if:
  - $S_0 \xrightarrow{A_0} (S_1, R_1) \xrightarrow{A_1} (S_2, R_2) \xrightarrow{A_2} (S_3, R_3) \cdots$
  - R_{n+1} is a deterministic or randomised function of (S_n, A_n, S_{n+1}) for all $n \geq 0$
  - Probabilistically, for all feasible trajectories of states, actions, and rewards,

$$\mathbb{P}\left(\{S_{n+1} = s', \ R_{n+1} = r\} \mid \{S_0 = s_0, \ A_0 = a_0, \ldots, S_n = s, \ A_n = a\}\right)$$

$$= \ \mathbb{P}\left(\{S_{n+1} = s', \ R_{n+1} = r\} \mid \{S_n = s, \ A_n = a\}\right).$$

  - The state S_n captures all information about the history $(S_0, A_0, R_1, \ldots, S_{n-1}, A_{n-1}, R_{n-1}, S_n)$
- In general, the choice of A_n could be based on the entire history up to time n

Time-Homogeneous MDP

- If the probability
$$\mathbb{P}\big(\{S_{n+1} = s',\ R_{n+1} = r\} \mid \{S_n = s,\ A_n = a\}\big)$$
is not a function of n, then the MDP is said to be **time-homogeneous**
- A state-reward dynamics in a time-homogeneous MDP may be specified by a **transition matrix**
$$P \ = \ \big\{P(s', r \mid s, a) : \ (s, a, s', r) \in \mathcal{S} \times \mathcal{A} \times \mathcal{S} \times \mathcal{R}\big\}.$$
- As usual, we have the stochastic structure
$$\sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} P(s', r \mid s, a) \ = \ 1 \qquad \forall\, (s, a) \in \mathcal{S} \times \mathcal{A}.$$

Time Homogeneous MDP
A time-homogeneous MDP is specified by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{R}, P)$.

S-A-S Transition Probabilities

- One can derive the following state-action-state transition probabilities:

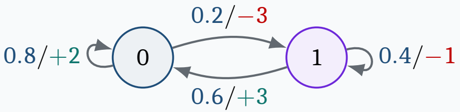
$$P(s' \mid s, a) \;:=\; \sum_{r \in \mathcal{R}} P(s', r \mid s, a), \qquad (s, a, s') \in \mathcal{S} \times \mathcal{A} \times \mathcal{S}$$

- S-A-S transition probabilities will play an important role in the design of RL algorithms

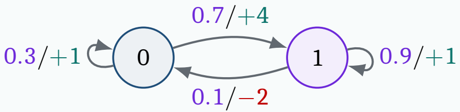
Example: A Two-State MDP

Edge label: transition probability / reward

Action a



Action b



The tuple $(\mathcal{S}, \mathcal{A}, \mathcal{R}, P)$

$$\mathcal{S} = \{0, 1\}, \qquad \mathcal{A}(0) = \mathcal{A}(1) = \mathcal{A} = \{a, b\}, \qquad \mathcal{R} = \{-3, -2, -1, 1, 2, 3, 4\}.$$

Example: The Recycling Robot

High charge: $\mathcal{A}(h) = \{\texttt{search}, \texttt{wait}\}$, **Low charge:** $\mathcal{A}(\ell) = \{\texttt{search}, \texttt{wait}, \texttt{recharge}\}$.
Let $\alpha, \beta \in [0, 1]$ and $r_s > r_w$.

State s	Action a	Next s'	$P(s' \mid s, a)$	$\mathbb{E}[R_{t+1} \mid s, a, s']$
h	`search`	h	α	r_s
h	`search`	ℓ	$1-\alpha$	r_s
h	`wait`	h	1	r_w
ℓ	`search`	h	$1-\beta$	-3
ℓ	`search`	ℓ	β	r_s
ℓ	`wait`	ℓ	1	r_w
ℓ	`recharge`	h	1	0

Why MDPs?
They help us model the interaction between an agent and its environment, where the agent's actions influence both the immediate rewards and the future states of the environment

Agent-Environment Interaction

At each time step n, the agent acts; the environment responds with feedback

Agent

observes history and chooses A_n

action A_n

reward R_{n+1}
next state S_{n+1}

Environment

$$(S_{n+1}, R_{n+1}) \sim P(\cdot \mid S_n, A_n)$$

The interaction loop $(S_0, A_0, R_1, \ldots, S_{n-1}, A_{n-1}, R_n, S_n) \to A_n \xrightarrow{P} (S_{n+1}, R_{n+1})$

Actions influence next state and reward; feedback informs future actions

Decision Rules and Policies

Decision Rule

- A decision rule represents the manner in which an agent chooses an action at a given time step n
- Let $H_n = (S_0, A_0, R_1, \ldots, S_{n-1}, A_{n-1}, R_n, S_n)$ denote the history up to time n
- Let $\mathcal{H}_n$ denote the set of all feasible histories H_n
  - A **history-dependent, deterministic** decision rule at time n is a function, say π_n, such that

$$\pi_n : \mathcal{H}_n \to \mathcal{A}, \qquad \pi_n : H_n \mapsto A_n \in \mathcal{A}.$$

  - A **history-dependent, randomised** decision rule at time n is a function, say π_n, such that

$$\pi_n : \mathcal{H}_n \to \Delta(\mathcal{A}), \qquad A_n \sim \pi_n(\cdot \mid H_n).$$

  - A **Markov, deterministic** decision rule at time n is a function, say π_n, such that

$$\pi_n : \mathcal{S} \to \mathcal{A}, \qquad \pi_n : S_n \mapsto A_n \in \mathcal{A}.$$

  - A **Markov, randomised** decision rule at time n is a function, say π_n, such that

$$\pi_n : \mathcal{S} \to \Delta(\mathcal{A}), \qquad A_n \sim \pi_n(\cdot \mid S_n).$$

Policy

- A policy of the agent is simply a collection of decision rules of the form $\pi = \{\pi_n\}_{n \geq 0}$
- Depending on the nature of the component decision rules, a policy may be categorised as:
  - **History-dependent, deterministic:** π_n is history-dependent and deterministic for each n
    - We will denote the set of all history-dependent, deterministic policies by Π^{HD}
  - **History-dependent, randomised:** π_n is history-dependent and randomised for each n
    - We will denote the set of all history-dependent, randomised policies by Π^{HR}
  - **Markov, deterministic:** π_n is Markov and deterministic for each n
    - We will denote the set of all Markov, deterministic policies by Π^{MD}
  - **Markov, randomised:** π_n is Markov and randomised for each n
    - We will denote the set of all Markov, randomised policies by Π^{MR}
- A Markov policy is said to be **stationary** if it uses the same decision rule at each time step n
  - A stationary, deterministic policy: $\pi_n = \varphi$ for all n, for some fixed $\varphi : \mathcal{S} \to \mathcal{A}$
  - Stationary, randomised policy: $\pi_n = \varphi$ for all n, for some fixed $\varphi : \mathcal{S} \to \Delta(\mathcal{A})$
- Π^{HR} is the most general class of policies that we will study in the course

Ways an Agent Can Choose Actions

A policy is classified by **what information it uses** and **how it selects an action**.

INFORMATION USED \ ACTION SELECTION	**Deterministic** one prescribed action	**Randomised** sample an action
H_n **History-dependent** full history H_n	**History-dependent, deterministic** $A_n = \pi_n(H_n)$	**History-dependent, randomised** $A_n \sim \pi_n(\cdot \mid H_n)$
S_n **Markov** current state S_n	**Markov, deterministic** $A_n = \pi_n(S_n)$	**Markov, randomised** $A_n \sim \pi_n(\cdot \mid S_n)$

$$H_n = (S_0, A_0, R_1, \ldots, S_{n-1}, A_{n-1}, R_n, S_n).$$

Suggested Reading

Markov Chains

1. **Sheldon M. Ross**
   Introduction to Probability Models, 12th ed., Academic Press, 2019.
   Chapter 4: Markov Chains

2. **J. R. Norris**
   Markov Chains, Cambridge University Press, 1997.
   Chapter 1: Discrete-time Markov chains

MDPs & Reinforcement Learning

1. **Martin L. Puterman and Timothy C. Y. Chan**
   Markov Decision Processes and Reinforcement Learning, Cambridge University Press, 2026.
   Chapter 2: Model Foundations, up to § 2.2.4

2. **Richard S. Sutton and Andrew G. Barto**
   Reinforcement Learning: An Introduction, 2nd ed., MIT Press, 2018.
   Chapter 1 and Section 3.1